

1. Introduction and scope

This report documents four generative adversarial network (GAN) experiments across three data modalities: two-dimensional point distributions, greyscale images, and tabular network-traffic records. Covering three modalities with one framework separates what belongs to the training procedure from what belongs to the data. Every model was implemented and trained by the author in PyTorch [19], and all figures and numeric values below are outputs of those runs.

Section 2 derives the equilibrium loss values used as a diagnostic throughout. Section 3 covers the from-scratch 2D GANs, where the target distribution can be plotted directly and correctness needs no proxy metric. Sections 4 to 6 cover a deep convolutional GAN (DCGAN) on retinal OCT scans, a fully connected GAN on the CICIDS 2017 intrusion-detection dataset, and a DCGAN on QuickDraw sketches, and Section 7 draws the four together. Two experiments contain outright failures; these are diagnosed rather than omitted, because in both cases the cause is traceable to a specific design decision and the diagnosis is the most informative result obtained.

2. The adversarial objective as a measuring instrument

A GAN couples a generator G , mapping noise z to a sample, against a discriminator D estimating the probability that a sample is real [1]. Both train simultaneously on opposed objectives: D minimises binary cross-entropy (BCE) against labels of one for real and zero for generated data, while G is updated to maximise D 's error on its own output. Because the objectives are opposed the losses are not expected to decrease; the system seeks a balance, not a minimum. That property is usually stated and then left alone, which makes the traces look uninterpretable. They are not. At the optimum the generated distribution equals the real one and D can do no better than chance, outputting 0.5 for every input, and substituting 0.5 into the BCE expression gives the exact equilibrium losses:

$$G \text{ loss} \rightarrow -\ln(0.5) = \ln 2 \approx 0.693$$

$$D \text{ loss} \rightarrow \ln 2 \approx 0.693 \text{ (mean over a combined real-and-fake batch)}$$

The parenthesis matters for reading the tables below. In Part 1 the discriminator is scored once on a concatenated batch of 32 real and 32 generated points, so its loss is a single mean and its equilibrium is $\ln 2 \approx 0.693$; in every Part 2 model it is the sum of two separate means, one on real and one on generated data, giving $2 \ln 2 \approx 1.386$. Each discriminator loss below is read against whichever constant applies, and distance from it measures how far the game is from balance.

2.1 Evaluation strategy per modality

No single metric works across all three modalities, so each experiment uses the strongest available evidence. Two-dimensional data is evaluated by plotting generated samples over the training data, which is the reason for working in 2D at all: correctness is directly visible, so no proxy metric is needed, and none can mislead. Images use Frechet Inception Distance (FID), comparing the mean and covariance of InceptionV3 features between real and generated sets, lower being closer [4], [20], paired with visual inspection because one number cannot distinguish a blurry sample from a structurally wrong one. Tabular data has no meaningful Inception embedding, so FID is inapplicable; both feature matrices are instead projected to two dimensions by PCA, which is linear and preserves global variance structure, and by t-SNE, which preserves local neighbourhoods [16]. The two are used together because each fail where the other works.

One caveat applies to every FID value below: they use 2,000 samples of each against 2,048-dimensional features, so each covariance estimate is rank-deficient, and FID is known to be biased at small sample sizes [21]. The values are comparable only with each other, all four having used an identical pipeline.

3. Part 1: two-dimensional GANs built from scratch

3.1 Architecture and its justification

Each sample is a single point (x, y) . The data has no spatial structure for a convolution to exploit, so both networks are multi-layer perceptrons; a convolutional layer would impose a neighbourhood relationship between the x and y coordinates that does not exist. The discriminator is 2-256-128-64-1 with ReLU, dropout of 0.3 after each hidden layer and a sigmoid output reading directly as a probability. The generator is 2-16-32-2 with ReLU and deliberately no output activation, since its output is a pair of unbounded coordinates that any squashing function would restrict.

The capacity asymmetry is severe: 41,985 discriminator parameters against the generator's 658, roughly 64 to 1. Dropout is therefore not a conventional overfitting control but a deliberate handicap, since a discriminator that becomes a perfect classifier saturates and the gradient reaching the generator vanishes, precisely the failure seen in Section 4.3. Training used Adam at learning rate 0.001 [17], batches of 32, 2,000 epochs and a fixed seed of 42. One structural point also conditions Task 1: the latent vector is two-dimensional, but the sine target is a one-dimensional curve in the plane, so the generator must learn a rank-deficient mapping collapsing one latent direction, and the boundary of the support is where it is least constrained.

3.2 Task 1: reproducing the sine-wave GAN

The training set is 1,024 points with x uniform on $[0, 2\pi]$ and $y = \sin(x)$. After 2,000 epochs the discriminator loss had settled into a band of 0.599 to 0.777 and the generator loss into 0.641 to 1.006, both centred close to the $\ln 2$ equilibrium. The game reached and held balance.

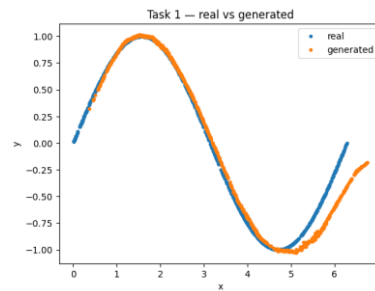


Figure 1: Task 1. One thousand generated samples (orange) over the 1,024-point sine training set (blue).

Figure 1 shows the generator tracing the curve accurately across the interior of its range. Two defects are visible and both concern the support rather than the shape: generated samples begin at roughly $x = 0.3$, leaving the leftmost segment unpopulated, and the generated support extends beyond $x = 2\pi \approx 6.28$ to approximately 6.8, where the generated branch sits near $y = -0.2$ while the real curve has already returned to $y = 0$. The generator learned the shape of the manifold but not its extent. This is consistent with the seed sensitivity the task asks about: the seed fixes both initial weights and the noise stream, and because balance is reached quickly, small differences in starting conditions are not later corrected, the endpoints being where that shows. Where a run instead concentrates on a subregion, the failure is mode collapse, the generator finding one area that reliably fools the discriminator and ceasing to explore.

3.3 Task 2: a noisy multi-frequency distribution

The second distribution is $y = \sin(2x) + 0.3\cos(5x) + \epsilon$, with ϵ normal of standard deviation 0.05 and x uniform on $[0, 2\pi]$. This is harder in two distinct ways: the cosine term adds a second, higher frequency, and the noise makes the support a genuine two-dimensional band rather than a curve, so spread must be modelled as well as shape. Architecture, optimiser, seed and epoch count were held identical to Task 1, making the data the only variable.

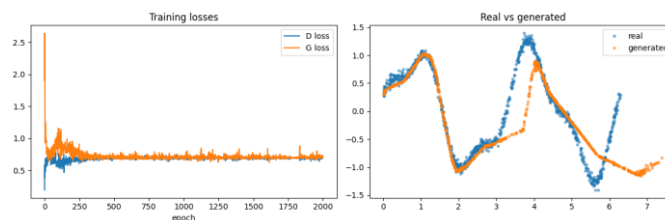


Figure 2: Task 2. Training losses over 2,000 epochs (left) and generated against real samples for the noisy curve (right).

The loss curves are the cleanest convergence in the study. After roughly 250 epochs both traces sit tightly around 0.70, essentially exactly the $\ln 2$ equilibrium, and hold there for the remaining 1,750 epochs. On loss evidence alone this run would be judged a complete success.

The sample plot contradicts that judgement, and the contradiction is the most useful finding in Part 1. The first lobe, x from 0 to about 2.5, is reproduced faithfully including the thickness of the noise band. Between $x = 3.0$ and 4.0, however, the real data dips to about $y = -0.5$ then climbs steeply to a peak near +1.35, whereas the generated samples cut a near-straight diagonal across that corner and reach only about +0.95, undershooting by roughly 0.4. The real trough near $x = 5.6$ reaches $y = -1.4$ while the generated samples bottom out near -1.15, and the final upswing between $x = 6.0$ and 6.28 is absent entirely, replaced by generated points extending to $x \approx 7.6$, outside the real support. Read together these say the generator learned the low-frequency $\sin(2x)$ component and smoothed over the higher-frequency $0.3\cos(5x)$ term. The methodological conclusion is that a GAN sitting precisely at its theoretical equilibrium can still be visibly wrong about the distribution: equilibrium indicates that neither network dominates but says nothing about whether the generator's distribution matches the target.

3.4 Task 3: architecture modification and comparison

Two changes were evaluated on the same data with the same seed. ReLU was replaced by LeakyReLU with negative slope 0.2 in both networks, so units receiving negative pre-activations still pass a gradient instead of switching off permanently; and the generator was deepened and widened from 2-16-32-2 to 2-32-64-32-2, raising its parameters from 658 to 4,354, a factor of 6.6. Discriminator width was unchanged, narrowing the capacity ratio from 64:1 to about 10:1.

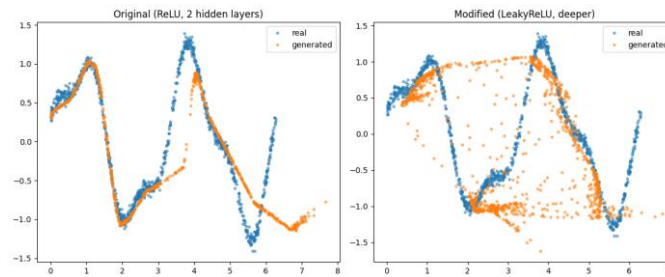


Figure 3: Task 3. Identical data and seed under the original architecture (left) and the modified deeper LeakyReLU architecture (right).

The comparison does not resolve into one architecture being better. The modified generator is clearly superior on coverage: it reaches the second peak at about $y = +1.05$ against the original's $+0.95$, populates the deep trough near $x = 5.6$ down to about -1.35 where the original stopped at -1.15 , and confines its support to roughly x in $[0.2, 6.6]$ instead of overshooting to 7.6 . It is equally clearly inferior on precision: a diagonal band of stray samples runs from about $(1.3, +1.0)$ to $(4.0, +0.9)$ through a region containing no real data, a horizontal band sits at $y \approx -1.05$ across x from 2 to 5 where the real distribution has no mass, and isolated outliers appear as low as -1.6 .

This is a precision-recall trade-off rather than a quality difference [7]. The original produces a thin, accurate, incomplete curve; the modified covers more of the true distribution while also placing mass where it has none. Judged by overall impression the original looks better, which is why both axes need reporting. The extra capacity was spent reaching further across the space rather than sharpening what had been found: with epoch budget and discriminator fixed, it bought recall at the cost of precision. Both runs held equilibrium throughout, the modified marginally more tightly at 0.680 to 0.710 , so the traces again failed to distinguish two visibly different outcomes.

4. Part 2.1: DCGAN on retinal OCT scans

OCTMNIST from MedMNIST v2 [12] provides 97,477 training scans at 28×28 greyscale across four diagnostic classes: normal 46,026 (47.2%), choroidal neovascularisation 33,484 (34.4%), diabetic macular oedema 10,213 (10.5%) and drusen 7,754 (8.0%), so the largest class is 5.9 times the smallest. Images were resized to 32×32 so three stride-2 halvings map exactly onto the resolution, and normalised to $[-1, 1]$ to match the generator's tanh output. Batches of 128 give 761 iterations per epoch.

4.1 Why a DCGAN, and why these specific choices

The Part 1 architecture would work on flattened images, but a fully connected layer treats each of the 1,024 pixels as independent and must relearn at every position that neighbouring pixels are correlated. The DCGAN design [2] builds that assumption into the weights. The generator upsamples through four transposed convolutions from 1×1 to 4×4 with 256 channels, then 8×8 , 16×16 and 32×32 with one channel, for 1.07 million parameters; the discriminator mirrors this with stride-2 convolutions to a single output, for 0.66 million. There are no pooling or fully connected layers, so strided convolutions let each network learn its own resampling.

Four further choices follow the DCGAN recipe, each for a reason. Batch normalisation after every hidden layer keeps activation scales stable while both networks move [18]. LeakyReLU at slope 0.2 in the discriminator ensures a gradient always reaches the generator, since a dead discriminator unit is an unrecoverable dead path for the generator's learning signal. Weights are initialised from a normal distribution of standard deviation 0.02, small enough that neither network saturates early. Adam runs at $2e-4$ with first momentum reduced from 0.9 to 0.5, because the gradient direction shifts as the opponent updates and heavy momentum carries the optimiser along directions that have stopped being descent directions.

4.2 Results after 30 epochs

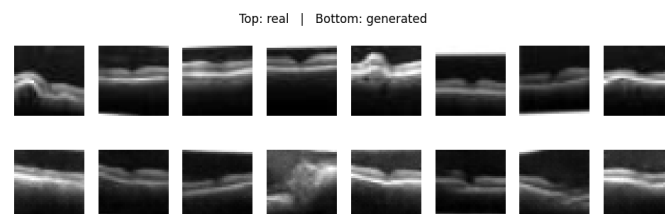


Figure 4: OCTMNIST. Real scans (top row) against DCGAN samples (bottom row).



Figure 5: OCTMNIST DCGAN. Sixty-four samples from a fixed noise batch (left) and per-epoch training losses (right).

FID over 2,000 real and 2,000 generated samples was 32.72. The samples in Figures 4 and 5 reproduce the defining structure of an OCT image: a bright hyper-reflective retinal band curving across a dark vitreous background, with layered banding beneath. Several show localised elevations of that band and several are near-flat, so variation in retinal profile has been captured rather than one template, and there is no mode collapse, the 64 samples differing in curvature, band thickness and brightness. Two defects are visible and both are attributable. A saturated bright stripe appears along the top or bottom edge in roughly a third of the samples: the characteristic edge artefact of transposed convolution, where uneven kernel overlap at the boundary produces periodic intensity spikes [8]. One of the eight samples in Figure 4 is also an incoherent blob. Across the grid the fine speckle texture of real OCT is softened, so macro-structure is reproduced more faithfully than micro-texture.

The loss traces tell a different story from Part 1. Against the applicable equilibrium of $2 \ln 2 \approx 1.386$, the discriminator loss averaged 0.238 across the 30 epochs and fell as low as 0.008, while the generator loss averaged 4.11 and peaked at 7.29. The discriminator dominated the entire run, and a generator loss of 5 implies it assigned the generator's output a realness probability of about $e^{-5} \approx 0.007$. Under the original saturating generator objective this would stall training, since the gradient of $\log(1 - D(G(z)))$ vanishes as $D(G(z))$ approaches zero. The implementation uses the non-saturating form instead, maximising $\log D(G(z))$, whose gradient grows rather than vanishes in that regime [1], [6]. This is why a badly unbalanced loss trace still produced usable samples, and it is the exact converse of Section 3.3: there, equilibrium coexisted with visible error; here, sustained imbalance coexisted with good output.

4.3 Extension: conditional GAN, and why it failed

The extension conditions generation on the diagnostic class so a chosen class can be requested on demand [3]. The generator concatenates a 50-dimensional learned label embedding to its 100-dimensional noise, giving 150 input channels. The discriminator receives the label as a separate 1,024-dimensional embedding reshaped into a full 32x32 second input channel. Anticipating the dominance seen in Section 4.2, three standard countermeasures were applied simultaneously: one-sided label smoothing with a real target of 0.9 [5], a reduced discriminator learning rate of $1e-4$ against the generator's $2e-4$, and two generator updates per discriminator update.

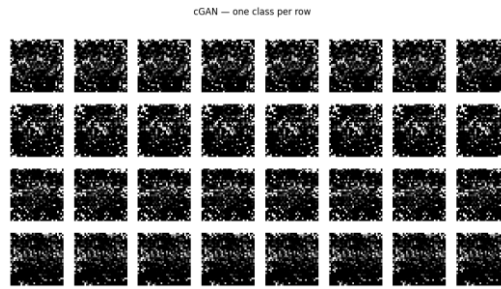


Figure 6: cGAN extension. Eight samples per row, each row requesting one diagnostic class. Within every row the eight samples are indistinguishable.

The model failed completely. Figure 6 contains no recognisable retinal structure, only salt-and-pepper noise with a faint horizontal band, and within each class row the eight samples generated from eight different noise vectors are visually identical. The generator collapsed to a single output per label, ignoring its noise input entirely: mode collapse in its most extreme form, one mode per class.

The training log identifies the cause precisely rather than by inference. The discriminator loss sat at 0.325 to 0.327 for 19 of the 30 epochs, and that value is not arbitrary. With a smoothed real target of 0.9, the BCE term on real data is minimised when the discriminator outputs exactly 0.9, and that minimum equals $-[0.9 \ln 0.9 + 0.1 \ln 0.1] = 0.32508$, while the term on generated data is minimised at zero. A total of 0.325 is therefore the exact global minimum of the discriminator's objective, not merely a low value: the discriminator had driven its own loss to the theoretical floor, meaning near-perfect separation of real from generated pairs. Over the same period the generator loss rose from about 7 to between 9 and 13, implying the discriminator scored its output between $e^{-9} \approx 1.2 \times 10^{-4}$ and $e^{-13} \approx 2.3 \times 10^{-6}$. The generator was operating where even the non-saturating objective gives no useful direction.

The mechanism is the conditioning channel. Supplying the label as a freely parameterised 1,024-dimensional embedding per class, reshaped to full image resolution, gives the discriminator 4,096 unconstrained parameters with which to learn a per-class spatial template. Real image-label pairs are consistent by construction, so any generator output not yet class-consistent is separable on that channel alone, before the generator has learned any image structure. The attribution rests

on a controlled comparison already in hand: the unconditional model of Section 4.2 shares the identical trunk, optimiser settings and data and succeeded, so the failure belongs to the conditioning mechanism rather than the backbone. Established remedies replace the free input channel with an inner product between the label embedding and the discriminator's features [9], constrain its Lipschitz constant by spectral normalisation [10], or move the label to a separate classification head [11]. The label smoothing merits a note: intended as a remedy, its contribution was diagnostic, establishing the 0.325 floor that made the saturation unambiguous.

5. Part 2.2: synthetic network traffic on CICIDS 2017

The core task uses the Wednesday capture of CICIDS 2017 [13], 692,703 flow records across 79 columns. The labels are severely imbalanced: BENIGN 440,031 (63.52%), DoS Hulk 231,073 (33.36%), DoS GoldenEye 10,293 (1.49%), DoS slowloris 5,796 (0.84%), DoS Slowhttptest 5,499 (0.79%) and Heartbleed 11 records (0.002%). Cleaning replaced infinities with missing values and dropped the affected rows, these arising from rate features whose denominator is a flow duration of zero, then retained numeric columns and discarded zero-variance ones, leaving 68 features. A random subsample of 20,000 rows was min-max scaled to [0, 1].

That subsample sets a hard ceiling worth stating before any result is interpreted. Sampling 20,000 of 692,703 records is a rate of 2.887%, so the expected number of Heartbleed flows in the training matrix is $11 \times 0.02887 = 0.32$. The rarest class is with high probability entirely absent from the data the generator ever sees, and no generator can synthesise a class not present in its training distribution.

5.1 Architecture and training behaviour

A flow record is an unordered feature vector: column 40 bears no metric relationship to column 41, so the weight sharing that makes convolution effective on images would impose a false neighbourhood structure. Both networks are therefore MLPs. The generator maps a 32-dimensional latent vector through 128 and 256 hidden units to 68 outputs, with batch normalisation and ReLU on the hidden layers and a sigmoid output matched to the [0, 1] min-max scaling, the same activation-to-scaling pairing used for tanh and $[-1, 1]$ on the images. The discriminator runs 68-256-128-1 with LeakyReLU and dropout of 0.3, load bearing here because sharply structured tabular records are easy to classify and an unregularised discriminator saturates quickly. Training used batches of 512, Adam at $2e-4$ with first momentum 0.5, and 150 epochs.

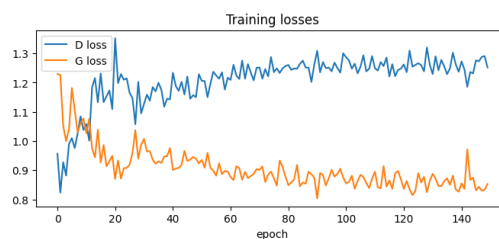


Figure 7: CICIDS GAN training losses across 150 epochs.

This is the best-behaved run in the study, and unlike Part 1 it converges towards equilibrium from outside rather than settling immediately. The discriminator loss rises from about 0.85 at the first epoch to a plateau near 1.25 to 1.30 while the generator loss falls from about 1.2 to about 0.85, both moving monotonically towards the applicable equilibrium pair of 1.386 and 0.693 and then holding. Final values were 1.252 and 0.853. Measured as distance from equilibrium this run is closer to a Nash equilibrium than any image model, whose discriminator losses sat below 0.2 against the same 1.386 target.

5.2 Comparison by PCA and t-SNE



Figure 8: CICIDS Wednesday. PCA fitted on real data and applied to both sets (left), and t-SNE on 2,000 samples of each (right).

The PCA projection reveals a strongly filamentary real geometry: a dense vertical spine at approximately $PC1 = -0.7$, with discrete horizontal rays extending to $PC1 \approx 2.7$ at distinct $PC2$ levels near $+0.55$, -0.28 , -0.65 and -0.95 . This reflects sets of flows identical in most features while varying along one, characteristic of automated traffic. The generated samples cover the spine well and do reproduce the rays, but with two errors: the generated rays terminate around $PC1 \approx 1.5$ to 2.0 rather than reaching 2.7 , truncating the tails of the distribution, and generated points fill the interior region between the rays where the real data has essentially no mass.

The second error is the more consequential, since those interior points are feature combinations no real flow exhibits, so synthetic records used to augment an intrusion-detection training set would inject invalid traffic. The same two-sided error

appears in the t-SNE view: most clusters contain both colours intermingled, but several towards the right and upper right are real-only while some regions are dense in generated points with little real support. One caution applies, in that t-SNE preserves local neighbourhoods but not global distances or densities [16], so the conclusion here rests only on which colours appear within a cluster, which the method does support.

5.3 Extension: all eight capture days

The extension combines all eight daily files, sampling up to 15,000 rows from each for 120,000 records across 13 attack labels. BENIGN accounts for 97,092 records, 80.9% of the total, and the rarest labels are Infiltration with 2 records and Web Attack SQL Injection with 3. A fresh model of the same architecture was trained on a 40,000-row subsample for 150 epochs.

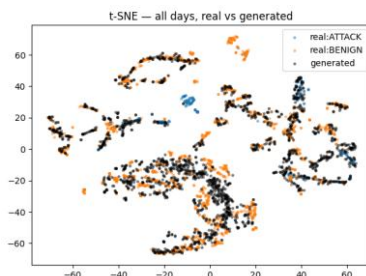


Figure 9: All eight days. t-SNE with real benign traffic, real attack traffic and generated samples overlaid.

Figure 9 is the clearest evidence in the study for selective mode coverage. The generated samples, in black, blanket almost the entire embedding and overlap the benign clusters extensively, but the compact cluster of real attack traffic near $(-8, 30)$ contains no generated points whatsoever: the generator covers the dominant benign traffic thoroughly and has not represented that attack mode at all. This follows from the objective rather than any implementation defect, since an unconditional GAN minimises a divergence between whole distributions and a mode carrying under one per cent of the mass contributes little to it, so omitting it is nearly free in loss terms. The traces cannot reveal this, which is why the projection was necessary. Addressing it means conditioning on the attack label with resampling so rare classes appear often enough to matter, as CTGAN does [14], or training separate models per attack family.

6. Part 2.3: DCGAN on QuickDraw sketches

The QuickDraw birthday-cake category [15] provides 144,982 bitmaps at 28x28. A subsample of 30,000 was resized to 32x32 and scaled to $[-1, 1]$, giving 234 iterations per epoch. Architecture, optimiser settings, resolution and epoch count are identical to Section 4.2, deliberately: with everything else constant, differences between this experiment and the OCT experiment are attributable to the data.

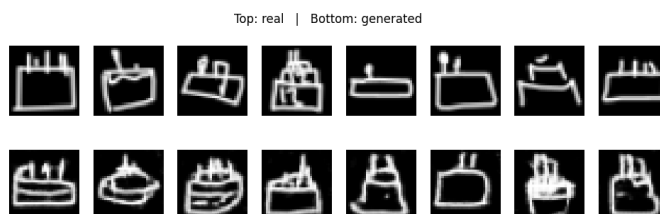


Figure 10: QuickDraw birthday cake. Real sketches (top row) against DCGAN samples (bottom row).

FID was 35.35, against 33.00 for OCT under the same pipeline. The generated sketches show a closed cake body with candle strokes rising from the top edge in many samples, with clear variation in body proportion, candle count and internal shading. Stroke quality is the weak point: a minority have broken or doubled contours, and a few are incomplete outlines. Loss behaviour matched the OCT run, the discriminator loss ranging from 0.017 to 2.218 and averaging well below the 1.386 equilibrium, so the same discriminator-dominant regime applies.

6.1 Extension: three categories under an identical pipeline

The same pipeline was run unchanged on two further categories. FID was 33.17 for house, 35.35 for cake and 108.38 for cat: houses and cakes are within 7% of each other, while cats are worse by a factor of more than three.



Figure 11: Generated cats, FID 108.38 (left) and generated houses, FID 33.17 (right), from an identical architecture and training budget.

Figure 11 shows what the metric is measuring. The house samples are near-uniformly a quadrilateral body beneath a triangular or pointed roof, frequently with a door or window, drawn in clean continuous single strokes on a clean black background. The cat samples are mostly recognisable frontal faces with pointed ears, many with eyes and whisker strokes, but a substantial minority are incoherent tangles with no identifiable structure. One further detail is visible only in the cat panel: the background carries grey speckle noise absent from both the house and cake panels, meaning the cat generator did not reliably learn even the constraint that background pixels are empty.

The explanation is intra-class variance rather than the complexity of any single drawing, and this study's own data rules out the complexity hypothesis: a house sketch contains more distinct strokes than a cat face, having a body, a roof and usually a door and windows, yet houses achieved the best FID. What distinguishes the categories is consistency of layout. House and cake are template-like, each following one dominant arrangement, so the generator learns a single structure well. QuickDraw cats are drawn as isolated faces, as full bodies, in profile and front-on, so the category is strongly multi-modal, and a fixed-capacity generator spread across many modes averages between them. This is the same mechanism as the uncovered attack cluster in Section 5.3 and the smoothed high-frequency term in Section 3.3.

7. Synthesis across the four experiments

Experiment	Data (size used)	Model (G / D params)	Epochs	D / G loss, last logged	Quantitative outcome
P1 Task 1	Sine wave, 1,024 pts	MLP 2-16-32-2	2,000	0.603 / 0.860	Curve tracked closely; support shifted right of the real range
P1 Task 2	Noisy curve, 1,024 pts	MLP 2-16-32-2	2,000	0.700 / 0.706	Low-frequency term learned; $\cos(5x)$ detail smoothed away
P1 Task 3	Noisy curve, 1,024 pts	MLP 2-32-64-32-2, LeakyReLU	2,000	0.692 / 0.685	Broader coverage of extremes, but off-manifold scatter
P2.1 core	OCTMNIST, 97,477 imgs	DCGAN, 1.07M / 0.66M	30	0.027 / 5.677	FID 33.00; layered retinal structure reproduced
P2.1 ext.	OCTMNIST, 97,477 imgs	cGAN, 1.07M / 0.67M	30	0.325 / 13.193	Failed: per-class collapse to a single noise pattern
P2.2 core	CICIDS Wed., 20,000 x 68	MLP 32-128-256-68	150	1.252 / 0.853	Dense core matched; distribution tails truncated
P2.2 ext.	CICIDS all days, 40,000 x 68	MLP 32-128-256-68	150	not logged	Isolated real attack cluster left entirely uncovered
P2.3 core	QuickDraw cake, 30,000	DCGAN, 1.07M / 0.66M	30	0.644 / 1.845	FID 35.35; varied, recognisable cakes
P2.3 ext.	QuickDraw house, 30,000	DCGAN, 1.07M / 0.66M	30	0.391 / 2.019	FID 33.17; best of the three categories
P2.3 ext.	QuickDraw cat, 30,000	DCGAN, 1.07M / 0.66M	30	0.613 / 2.341	FID 108.38; 3x worse, visibly incoherent samples

Table 1: All runs. The equilibrium reference for discriminator loss is $\ln 2 \approx 0.693$ for the Part 1 models, which score one mean over a combined batch, and $2 \ln 2 \approx 1.386$ for all Part 2 models, which sum two separate means. FID values are comparable within this study only, per Section 2.1.

7.1 What the equilibrium diagnostic did and did not reveal

Ranking the four by distance from their applicable equilibrium puts the CICIDS model closest, at 0.134 below 1.386 and still converging; the Part 1 models next, essentially on $\ln 2$; the two DCGANs far off, near 0.24 against a 1.386 target; and the conditional GAN pinned at its 0.325 floor. That ranking does not correspond to sample quality, and the extreme cases contradict it in opposite directions: the Task 2 model held equilibrium almost perfectly for 1,750 epochs while visibly failing to reproduce a peak, a trough and the end of its support, whereas the OCT DCGAN ran far from equilibrium and produced convincing scans at FID 32.72. The diagnostic detects training pathology, and the conditional GAN's 0.325 floor is a case where the loss trace alone identified both a failure and its mechanism, but it does not measure how close the generated distribution is to the target. Every quality conclusion here rests on a projection, an overlay or FID.

7.2 What determined success

- The mode structure of the data dominated every architectural choice made. The same DCGAN, at the same capacity and epoch budget, scored 33.17 on houses and 108.38 on cats. Consistent, template-like distributions were learned well, and strongly multi-modal ones were not, and this single factor explains the cat result, the uncovered CICIDS attack cluster and the smoothed high-frequency term in Part 1.
- The output activation must match the data scaling. Tanh with $[-1, 1]$ on images and sigmoid with $[0, 1]$ min-max scaling on tabular data is the same decision applied twice; getting it wrong places part of the generator's reachable range permanently outside the data's range.

- Conditioning must not hand the discriminator a free discriminative channel. A 1,024-parameter-per-class embedding supplied at full image resolution let the discriminator reach its exact loss floor before the generator had learned any image structure, defeating three simultaneous countermeasures, while the same trunk without that channel trained successfully.
- Additional generator capacity bought coverage, not accuracy. The 6.6x larger generator in Task 3 reached extremes the smaller one missed while also placing mass in empty regions, so capacity should be increased with an explicit position on which side of the precision-recall trade-off is wanted [7].
- The framework is domain-agnostic. One unmodified DCGAN implementation produced retinal pathology scans, birthday cakes, cats and houses, with only the training file changing. Subject matter is a property of the data, never the architecture, and a model can only generate what its training distribution contains, which is why the Heartbleed class of Section 5 was unreachable.

7.3 Limitations and further work

The image models ran for 30 epochs at 32x32, bounding achievable texture detail and contributing to the softening noted in Sections 4.2 and 6. FID rests on rank-deficient covariance estimates and is not comparable with published results [21]. Part 1 uses a single seed per configuration, so the Task 3 comparison establishes a difference between two runs rather than a distribution over runs, and the CICIDS work uses a 2.887% subsample excluding the rarest class in expectation. The largest gap is that the conditional experiment was diagnosed but not repaired; the projection-discriminator and spectral-normalisation remedies of Section 4.3 are the natural next step, with the unconditional baseline at FID 32.72 as the reference for any repair.

8. Conclusion

Four GANs were implemented and evaluated across 2D point distributions, greyscale images and tabular network traffic. The 2D models reached their theoretical equilibrium and reproduced their target curves with characterised errors at the extremes of support; the OCT DCGAN reached FID 32.72 with convincing layered retinal structure and no mode collapse; the QuickDraw DCGAN reached FID 35.35 on cakes and 33.17 on houses. The tabular GAN converged closest of the four to a Nash equilibrium and matched the dense core of the traffic distribution while truncating its tails and generating invalid feature combinations. The conditional GAN failed completely, the training log localising the cause to the discriminator reaching the exact global minimum of its smoothed objective, 0.32508, driven by a conditioning channel supplying 4,096 free parameters of per-class spatial template.

The recurring finding is that the mode structure of the data governed outcomes more strongly than any architectural decision taken: an identical architecture varied threefold in FID between two sketch categories of comparable per-drawing complexity but very different internal consistency, and the same mechanism accounts for the missed attack cluster and the smoothed frequency component in 2D. The secondary finding is methodological. Adversarial losses, read against their derived equilibrium values, are a precise instrument for detecting training pathology and an unreliable one for judging sample quality; every quality conclusion here came from a projection, an overlay or FID.

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